

TURKMENBASHI, Turkmenistan

WMO: 381412

Lat: 40.063N

Lon: 53.007E

Elev: 279

StdP: 14.55

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.4	22.9	4.6	7.0	28.7	10.4	9.4	34.6	25.8	40.7	23.4	42.4	9.0	80	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.0	102.0	67.4	99.1	66.9	96.8	66.7	77.0	87.8	75.0	86.3	73.3	85.3	12.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.8	127.1	82.7	71.6	117.9	81.1	69.6	109.9	80.2	40.7	87.6	38.7	86.2	37.1	85.4	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.8	21.7	19.7	DB	12.4	106.1	7.6	2.6	6.9	107.9	2.5	109.4	-1.8	110.9	-7.3	112.8
			WB	11.3	80.2	7.3	2.9	6.0	82.3	1.7	84.1	-2.4	85.7	-7.7	87.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.3	38.8	39.7	47.4	57.2	68.8	78.3	83.7	84.1	73.8	60.8	48.9	41.3	(6)
	DBStd	17.73	7.05	7.74	6.63	7.24	6.55	6.43	4.98	5.05	5.98	7.00	7.39	6.92	(7)
	HDD50	1179	352	293	129	17	0	0	0	0	0	8	102	278	(8)
	HDD65	3744	814	708	547	250	35	1	0	0	0	166	483	734	(9)
	CDD50	4958	3	5	48	234	583	848	1044	1057	715	342	70	9	(10)
	CDD65	2048	0	0	1	17	153	399	579	592	271	35	1	0	(11)
	CDH74	25244	0	0	11	192	1566	4821	7650	8009	2712	279	4	0	(12)
	CDH80	13164	0	0	0	37	553	2494	4318	4605	1120	37	0	0	(13)
(14) Wind	WSAvg	9.2	9.1	9.2	9.7	9.1	9.1	10.4	10.3	9.8	9.2	8.1	8.0	8.7	(14)
(15) Precipitation	PrecAvg	5.3	0.5	0.5	0.8	0.6	0.5	0.1	0.1	0.1	0.1	0.5	0.8	0.6	(15)
	PrecMax	9.5	1.3	1.9	3.5	2.0	5.6	1.0	0.7	1.1	1.1	2.2	3.4	1.9	(16)
	PrecMin	2.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	1.7	0.3	0.4	0.7	0.5	0.8	0.2	0.2	0.2	0.2	0.5	0.7	0.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.7	63.0	75.1	84.5	95.2	102.6	105.4	105.6	98.5	84.1	73.0	62.8	(19)
		MCWB	47.0	48.5	55.0	58.8	63.1	66.4	67.7	68.0	64.8	58.6	57.0	49.7	(20)
	2%	DB	53.8	57.2	68.0	79.1	89.7	98.9	101.8	102.1	93.2	80.3	67.1	57.5	(21)
		MCWB	45.4	46.3	51.2	57.9	61.9	66.2	68.5	66.9	63.5	59.2	54.8	48.4	(22)
	5%	DB	50.5	53.8	63.0	75.3	86.0	95.4	98.9	99.2	89.7	77.1	63.6	53.9	(23)
		MCWB	44.1	45.3	50.2	56.5	61.4	65.9	68.1	66.3	63.5	59.3	53.9	47.2	(24)
	10%	DB	48.0	50.4	58.9	71.2	82.3	92.3	96.3	96.9	86.3	73.7	60.5	51.5	(25)
		MCWB	43.3	43.7	47.9	55.0	61.0	65.9	68.1	66.3	63.2	58.8	52.8	46.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.5	50.3	56.5	62.1	68.6	77.4	79.7	79.6	75.1	69.4	61.3	53.1	(27)
		MCDB	54.0	58.5	69.8	77.7	82.9	89.9	90.2	90.5	84.7	75.3	68.0	60.4	(28)
	2%	WB	47.0	48.1	53.1	60.1	66.3	73.0	77.3	76.9	71.9	65.9	57.8	50.3	(29)
		MCDB	51.6	55.2	64.2	74.5	79.8	86.4	89.0	86.4	82.2	73.0	63.7	54.7	(30)
	5%	WB	45.1	46.2	51.0	58.3	65.0	70.8	75.2	75.0	69.8	63.5	55.7	48.5	(31)
		MCDB	48.7	51.3	60.6	71.1	78.2	83.9	87.3	85.6	80.6	70.9	61.2	52.8	(32)
	10%	WB	43.6	44.5	49.1	56.6	63.6	69.2	73.5	73.2	67.8	61.5	53.7	46.5	(33)
		MCDB	47.1	49.3	57.1	67.9	76.6	82.7	85.5	85.4	78.9	69.3	59.0	50.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.0	15.1	16.5	18.7	20.3	20.9	20.1	21.0	21.0	19.3	15.8	13.5	(35)
		MCDBR	15.4	20.3	23.7	25.1	25.2	24.8	23.9	24.6	24.4	23.0	20.1	16.0	(36)
		MCWBR	10.1	11.9	11.7	10.8	10.2	10.4	10.0	11.1	12.3	12.3	12.0	10.6	(37)
	5% WB	MCDBR	13.3	16.7	20.4	22.1	21.1	20.8	19.3	19.8	20.4	18.4	16.3	13.7	(38)
		MCWBR	9.7	10.9	10.9	10.5	9.7	10.8	10.6	11.4	12.1	12.4	11.2	10.2	(39)
(40) Clear-Sky Solar Irradiance	taub		0.412	0.454	0.490	0.568	0.547	0.540	0.576	0.542	0.520	0.488	0.440	0.421	(40)
	taud		2.071	1.972	1.921	1.781	1.875	1.912	1.842	1.907	1.921	1.967	2.042	2.049	(41)
	Ebn at Noon		223	232	239	229	239	240	230	233	228	218	211	207	(42)
	Edh at Noon		37	47	55	68	63	61	65	59	55	47	38	36	(43)
(44) All-Sky Solar Radiation	RadAvg		585	868	1254	1642	2090	2374	2239	2064	1686	1124	721	526	(44)
	RadStd		59	78	94	174	100	58	103	91	69	91	50	49	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.43	N/A	N/A	+2.03	N/A	N/A	N/A	N/A	+221	+234
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page